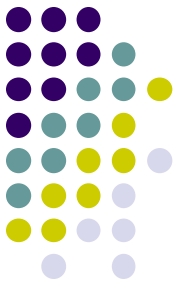
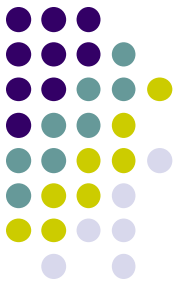


UML Activity Diagrams



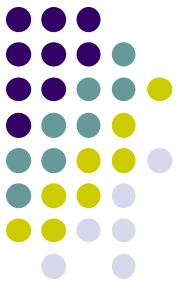
Introduction

- one of the five diagrams in the UML for modeling the dynamic aspects of systems
- An activity diagram is essentially a flowchart, showing flow of control from activity to activity. Unlike a traditional flowchart, an activity diagram shows concurrency as well as branches of control.
- Activity diagrams represent the business and operational workflows of a system



Definitions

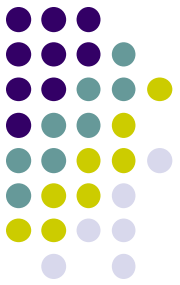
- A diagram that shows the flow of control and data from activity to activity.
- An activity is a behavior expressed as a set of actions connected by control and data flows.
- An action is a executable computation that results in a change in state of the system or the return of a value.



Elements

- Initial Activity
- Final Activity
- Activity
- Branching
- Synchronization Bar
- Flows (transitions)
- Swimlanes

Notations



Activity



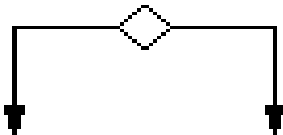
Initial State



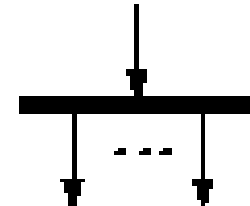
Final State



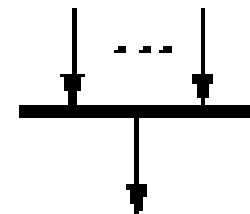
Transition



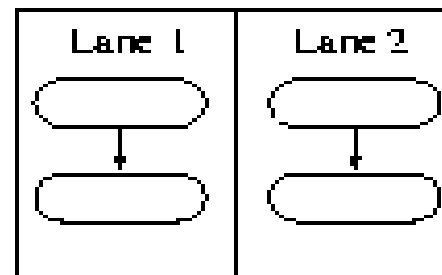
Branching



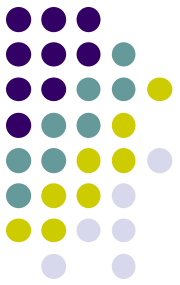
Fork



Join



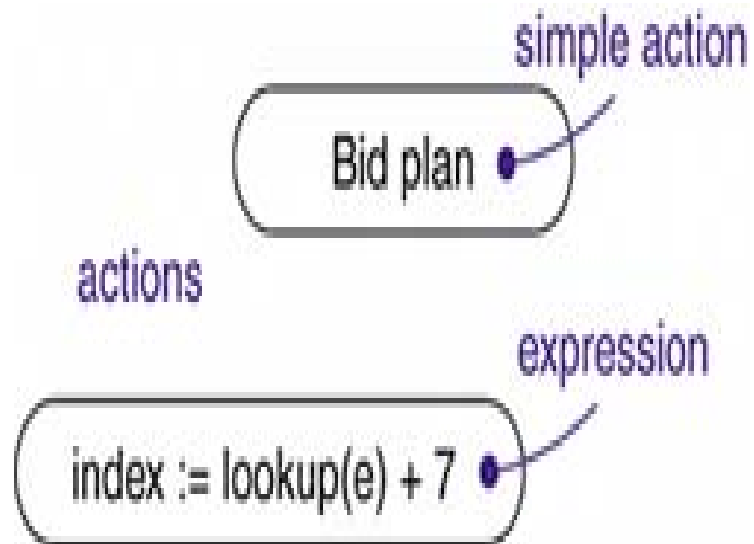
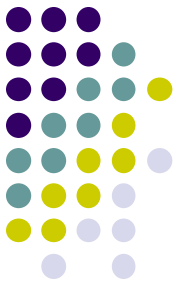
Swimlanes

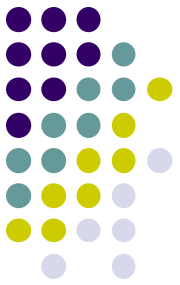


Actions and Activity Nodes

- You might evaluate some expression that sets the value of an attribute or that returns some value.
- you might call an operation on an object,
- send a signal to an object,
- or even create or destroy an object.
- These executable, atomic computations are called **actions**.

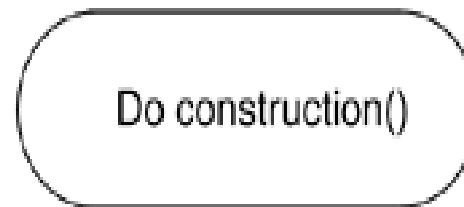
Actions

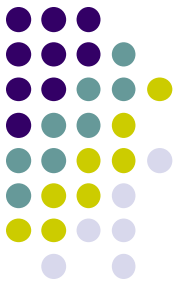




Activity Nodes

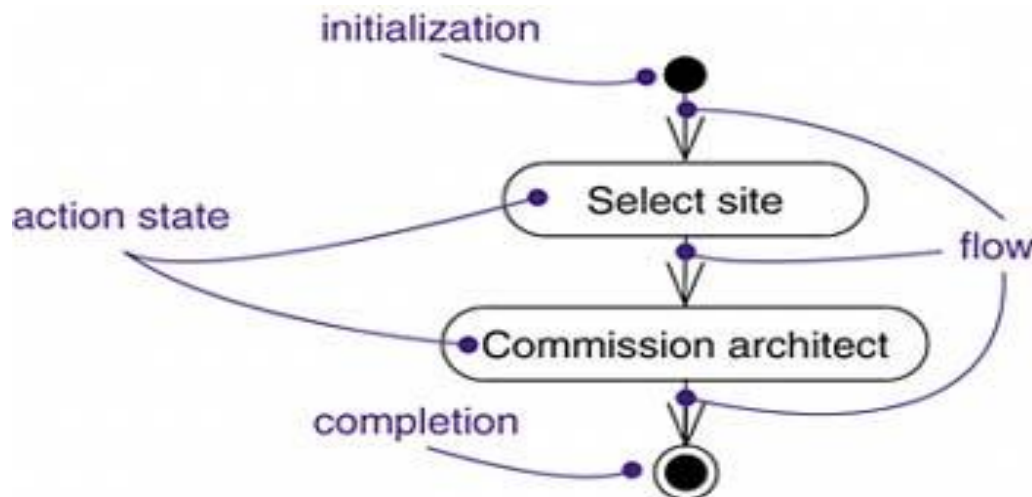
- An activity node is an organizational unit within an activity.
- activity nodes are nested groupings of actions or other nested activity nodes.
- An action is an activity node that cannot be further decomposed.

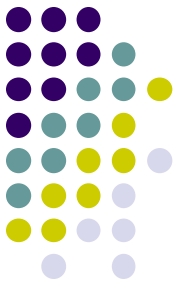




Control Flows

- When an action or activity node completes execution, flow of control passes immediately to the next action or activity node.
- use **flow arrows** to show the path of control from one activity to another activity

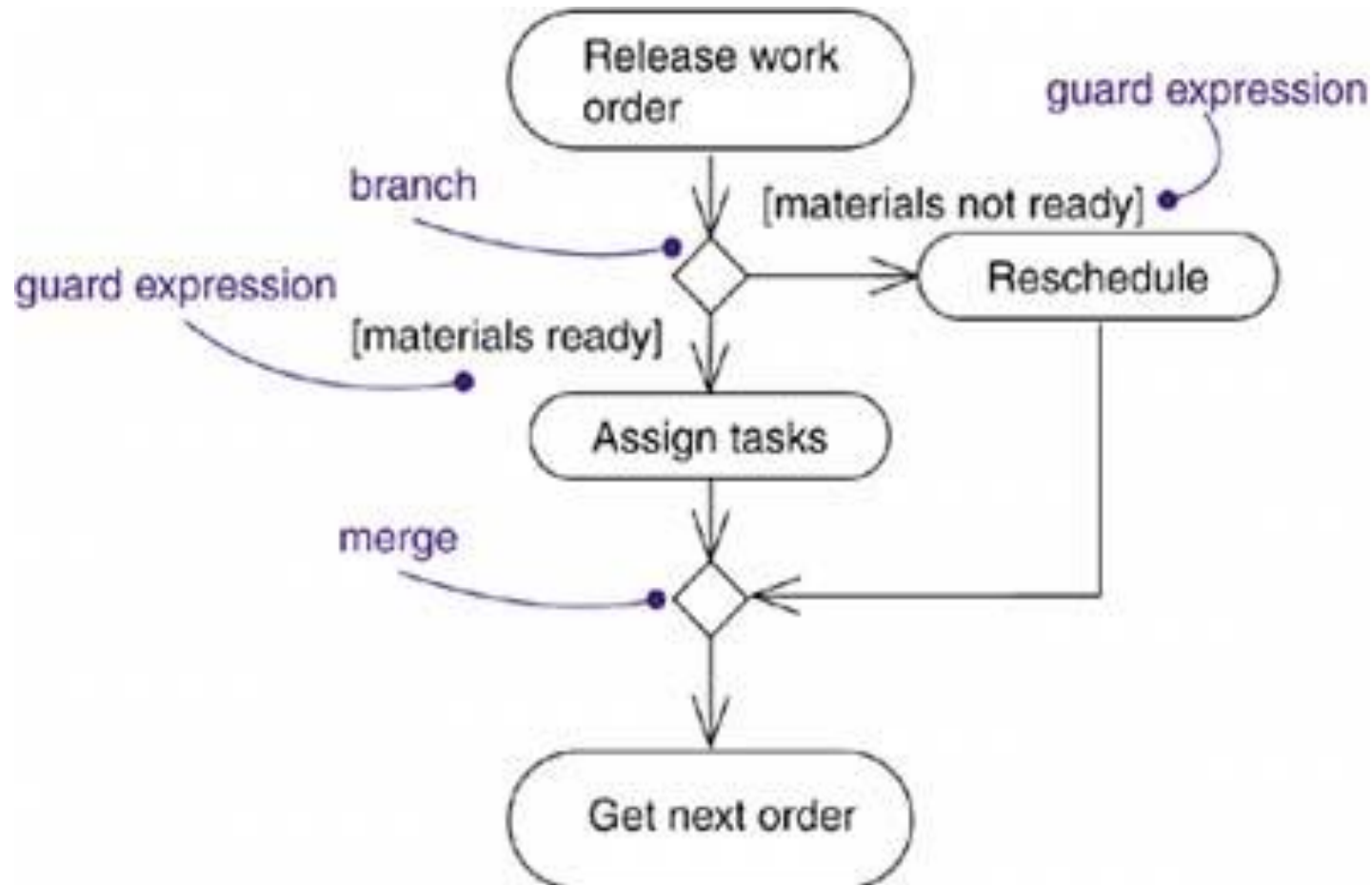
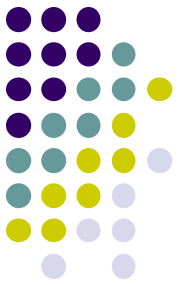


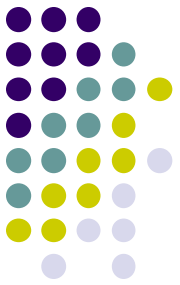


Branching

- specifies **alternate paths** taken based on some Boolean expression.
- represent a branch as a **diamond**.
- A branch may have **one incoming** and **two or more** outgoing flows.
- On each outgoing flow, you place a **Boolean expression**, which is evaluated

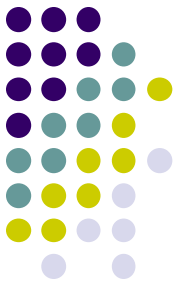
Branching





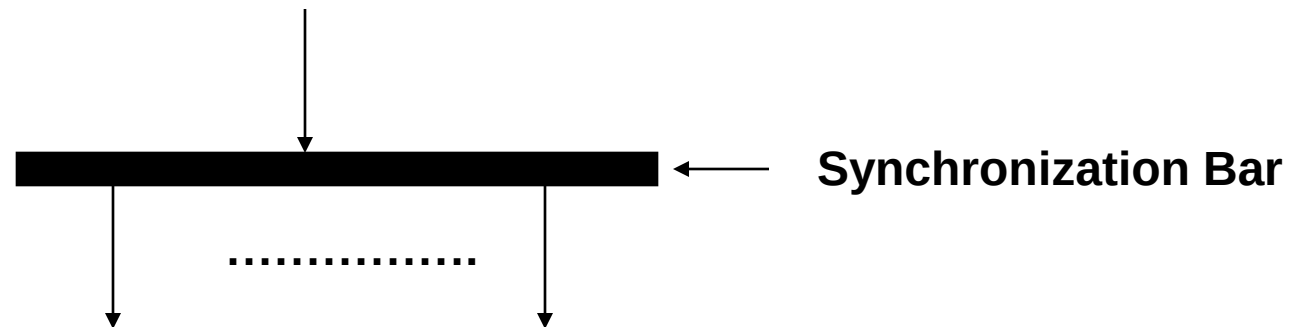
Forking and Joining

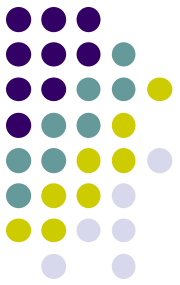
- Specify flows that are **concurrent** (activities in parallel)
- For example, **withdrawal and account updation** is a parallel activity.
- use a **synchronization bar** to specify the forking and joining of these parallel flows of control.
- A synchronization bar is rendered as a **thick horizontal or vertical line**.



Fork

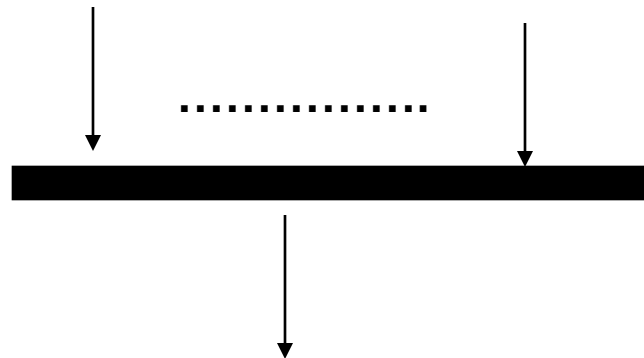
- a **fork** represents the **splitting of a single flow** of control into **two or more concurrent** flows of control.
- fork may have **one incoming transition** and **two or more outgoing transitions**

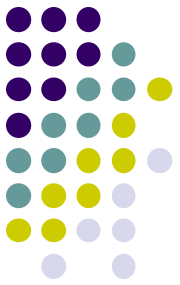




Join

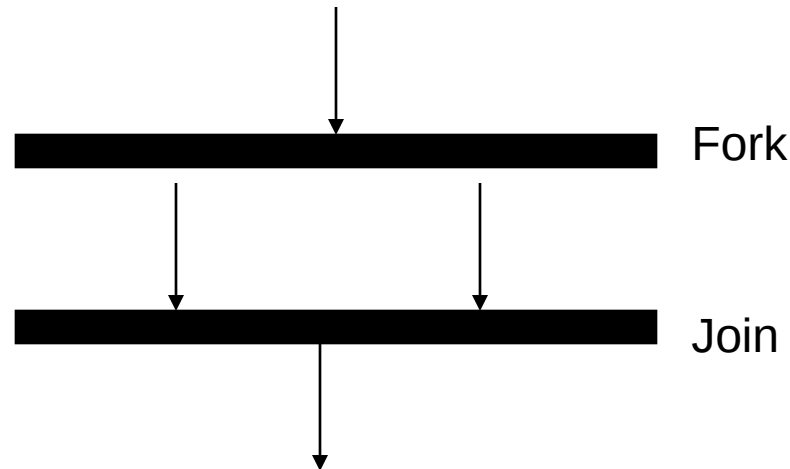
- **join** represents the **synchronization** of two or more concurrent flows of control
- may have **two or more incoming** transitions and **one outgoing** transition

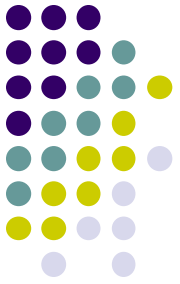




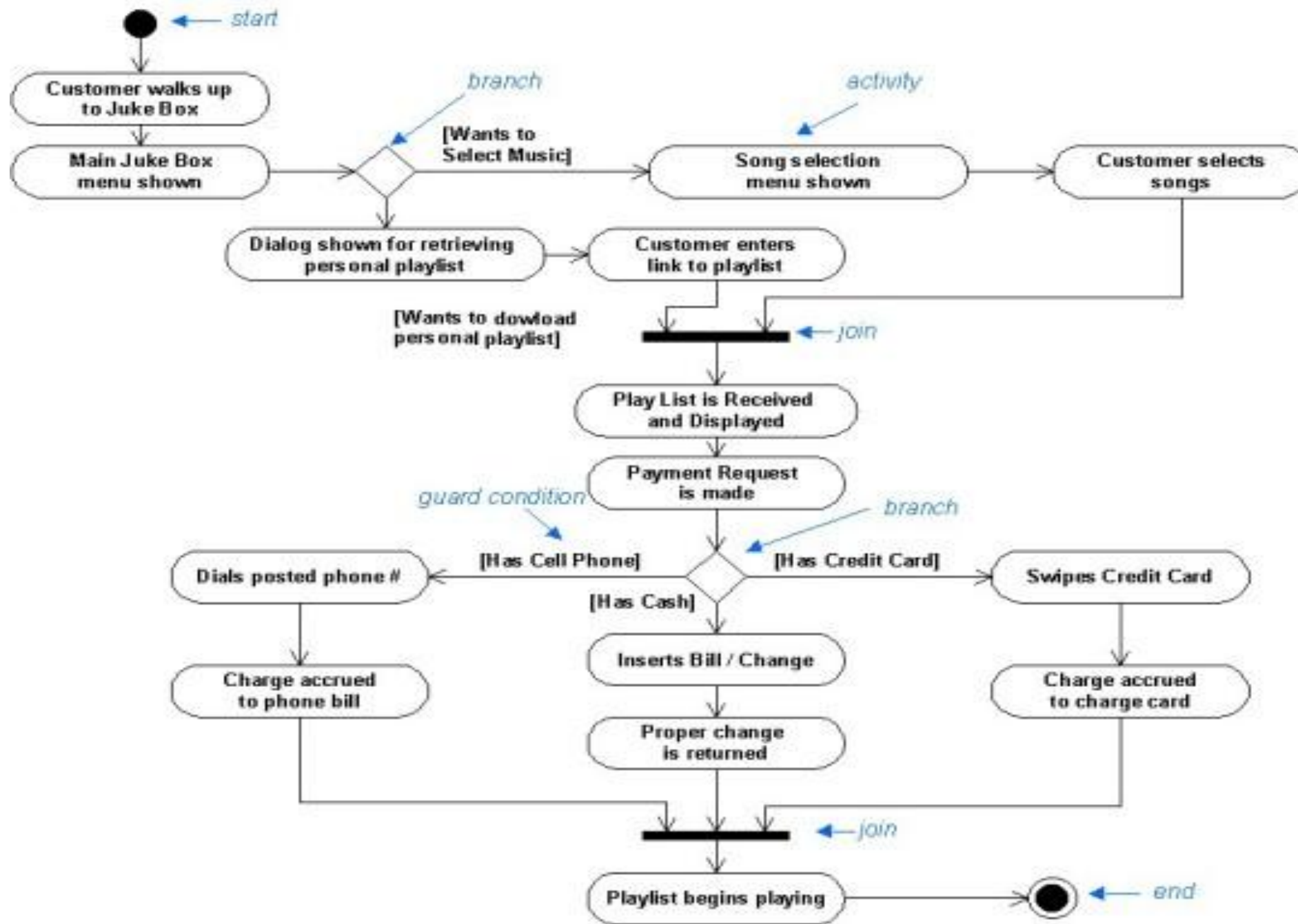
Note for fork and join

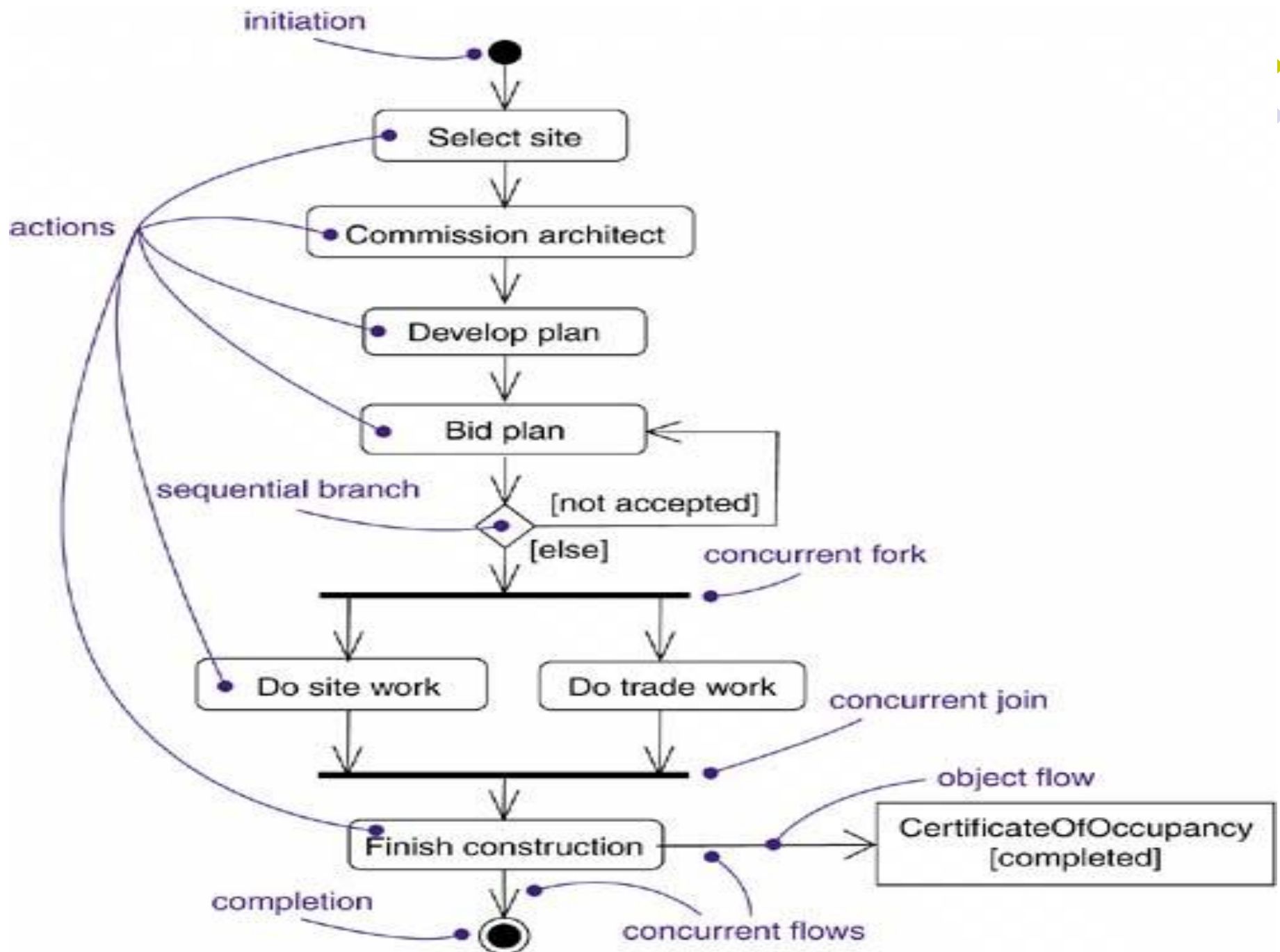
- Joins and forks should **balance**, meaning that the **number of flows that leave** a fork should match the **number of flows that enter** its corresponding join



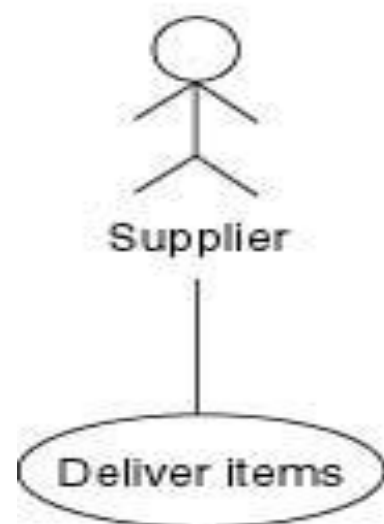


Examples

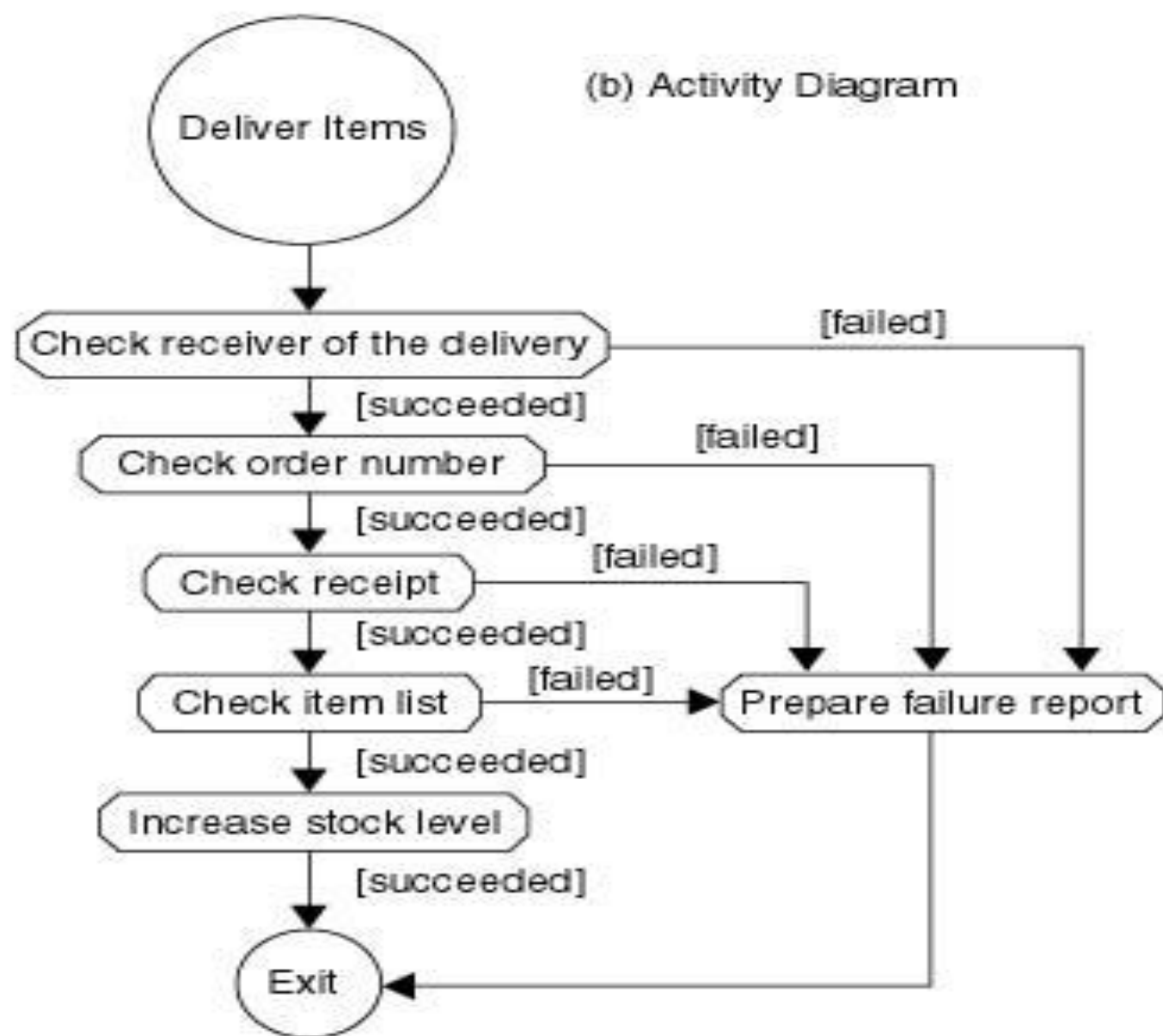




(a) Use case

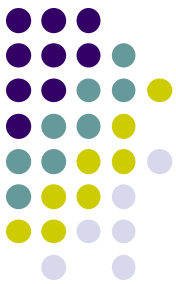
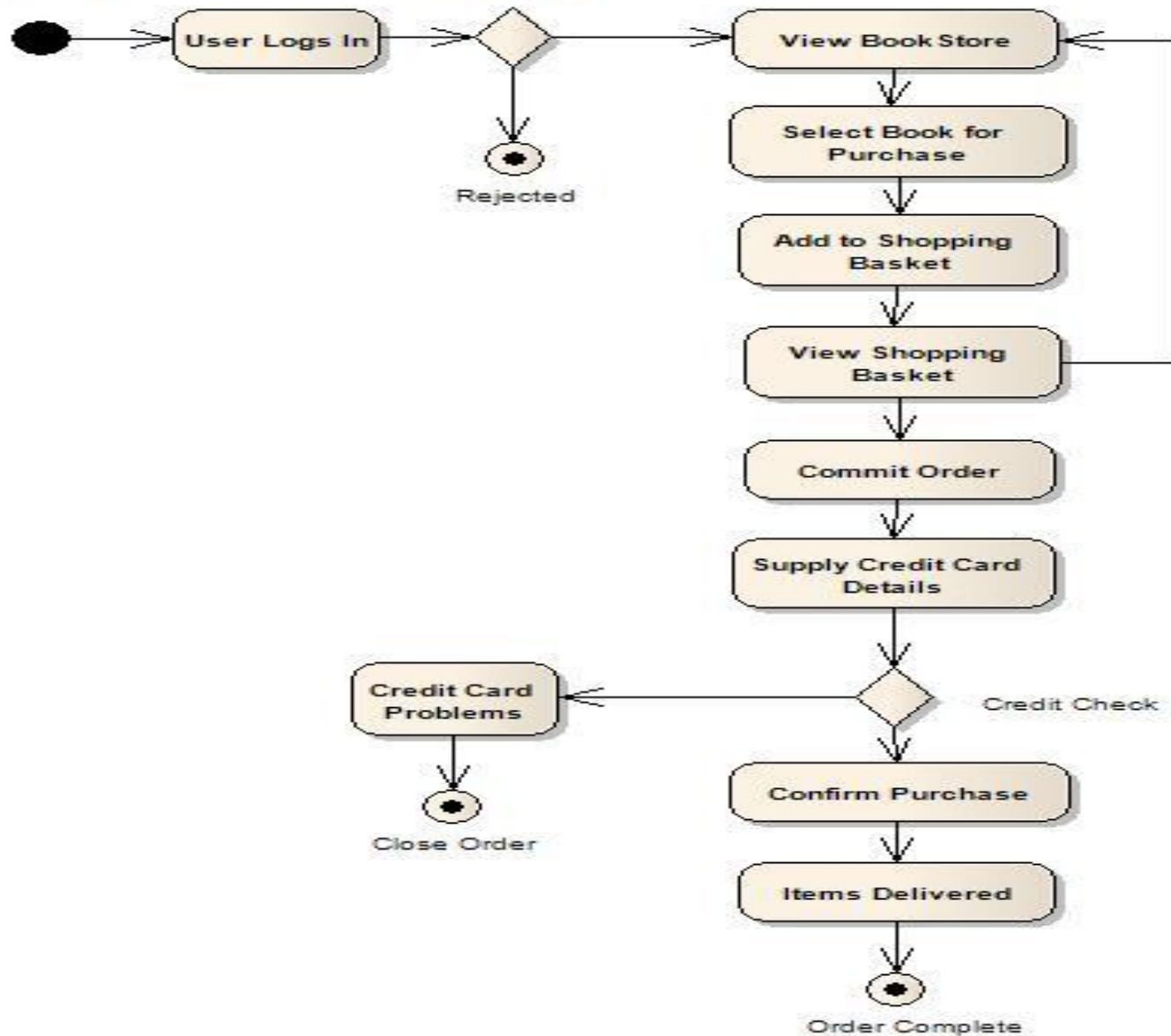


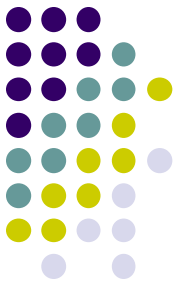
(b) Activity Diagram



act Customer Process

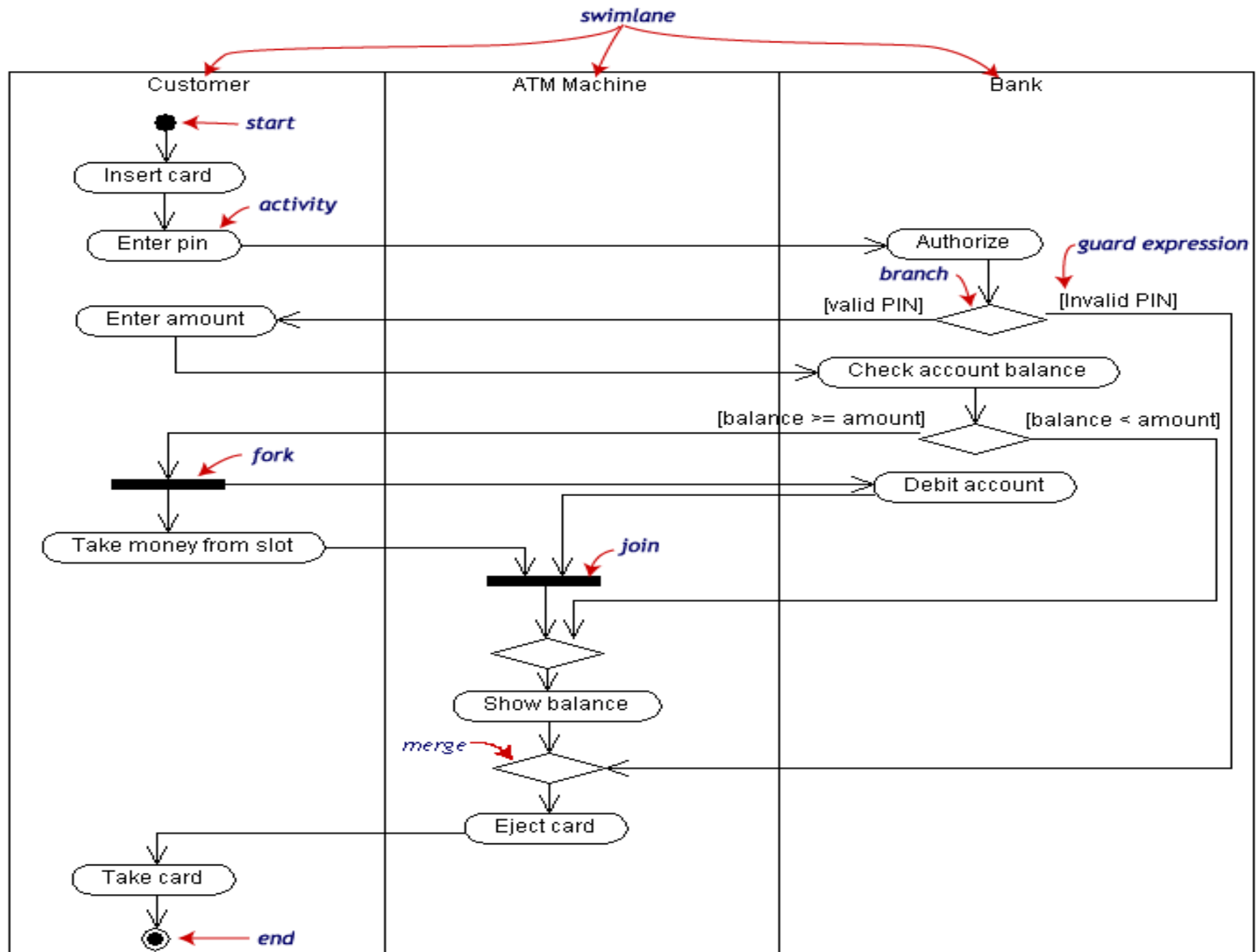
Customer Enters Web site





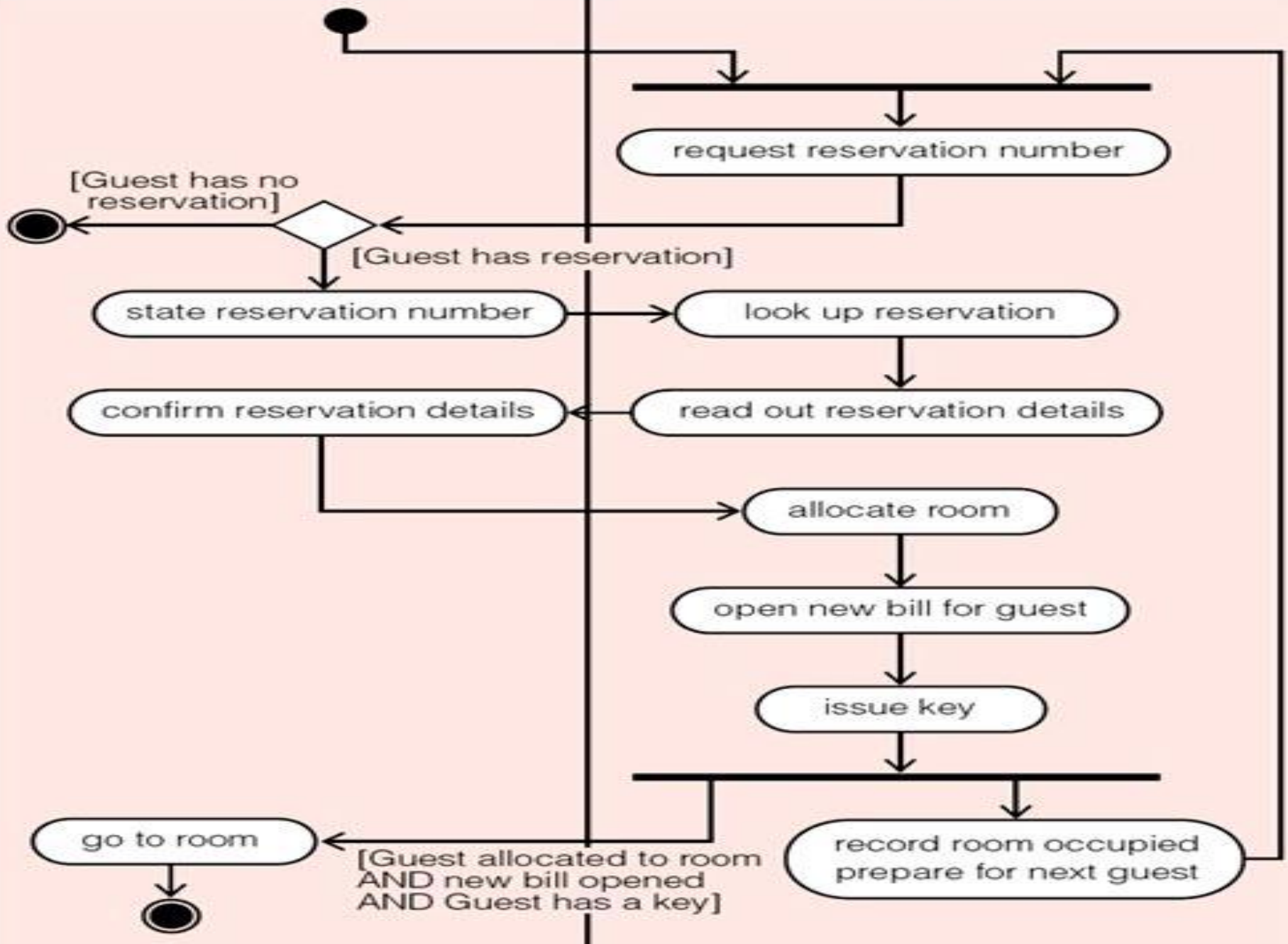
Swimlanes

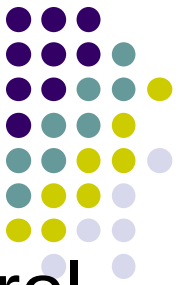
- Specify **which actor** performs which **activity**.
- specifies a **set of activities** that share some organizational property.
- Each swimlane **has a name** unique within its diagram
- In an activity diagram partitioned into swimlanes, **every activity** belongs to **exactly one swimlane**, but **transitions** may **cross lanes**.
- **A swimlane is a kind of package**
- Represented as thick **vertical solid line**



Guest

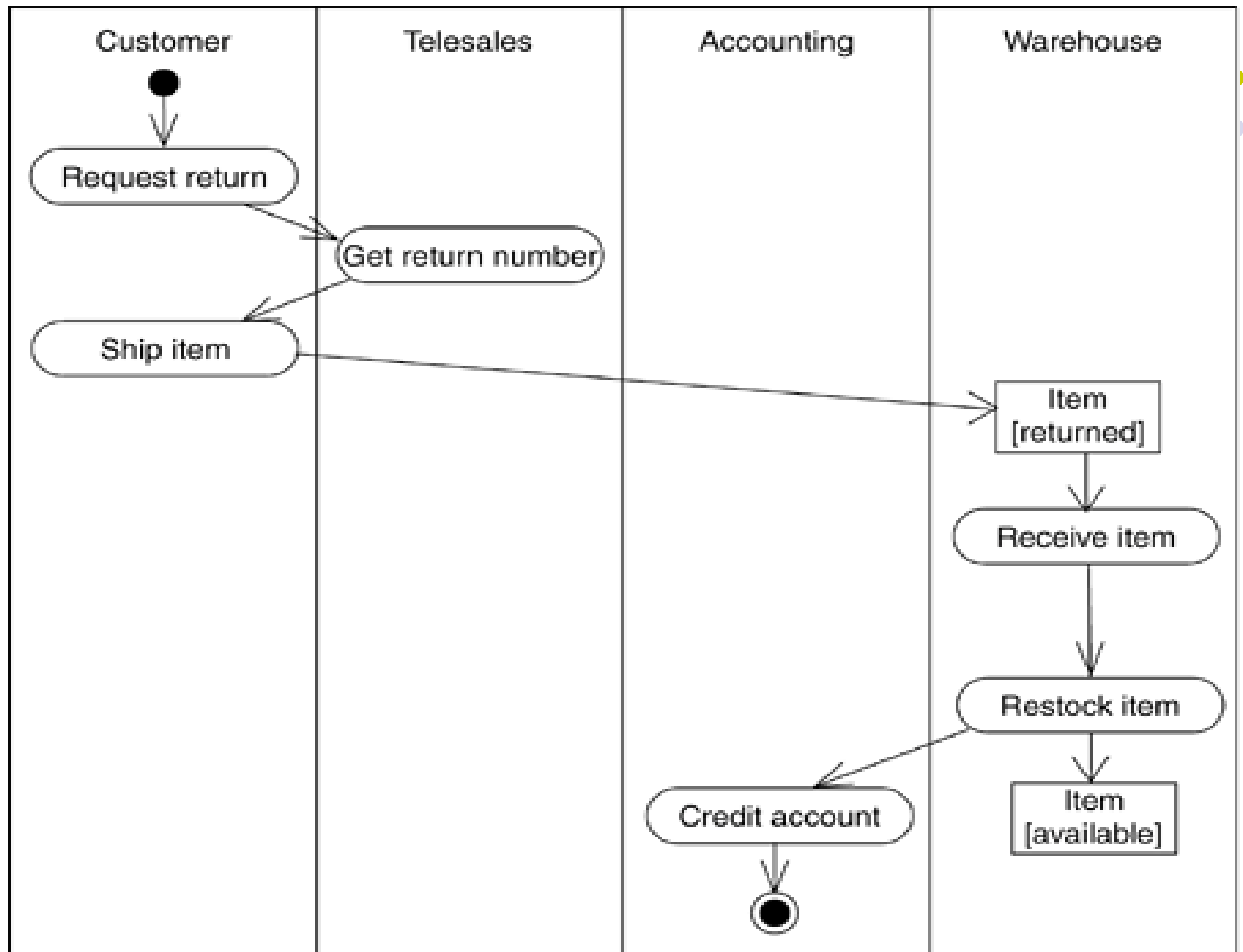
Receptionist





Object Flow

- Objects may be involved in the flow of control associated with an activity diagram
- Place these **objects** in the diagram, **connected by arrows** to the **actions** that produce or consume them.
- represents the **flow of an object value** from one action to another.
- you can also **show** how its **state changes**
- you represent the state of an object by naming its **state in brackets** below the object's name.

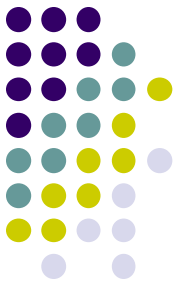


Exercise 1



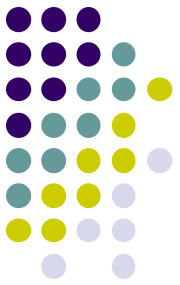
- A customer decides to upgrade his/her PC and purchase a DVD player. Customer begins by calling the sales department of PC vendor and they tell him/her to talk to customer support. He/She then calls customer support and they put him/her on hold while checking various DVD options. Finally, customer support tells the customer about several supported DVD options. The customer chooses a DVD and the mail department ships it. The customer receives the DVD, installs it satisfactorily, and then mails her payment to accounting.
- Construct an activity diagram for above scenario. Use swimlanes to show various interactions.

Exercise 2



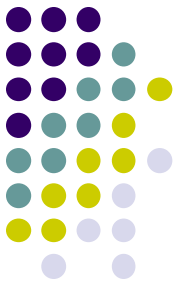
- Construct the activity diagram for activities related to eating in an Restaurant. Clearly specify Swimlanes.

Exercise 3



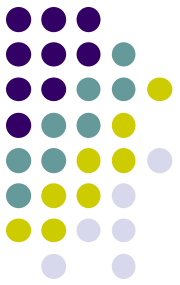
Read the following scenario:

- The ViaNet bank client must be able to deposit an amount to and withdraw an amount from his or her accounts using the bank application. Each transaction must be recorded, and the client must have the ability to review all transaction performed against a given account. Recorded transactions must include the date, time, transaction type, amount , and account balance after the transaction.
- A ViaNet bank client can have two types of accounts: a checking account and a savings account. For each checking account, one related saving account can exists.



Contd...

- The application must verify that a client can gain access to his or her account by identification via personal identification number (PIN) code.
- Neither a checking account nor a savings account can have a negative balance. The application should automatically withdraw funds from a related savings account if the requested withdrawal amount on checking account is more than its current balance.
- If the savings account balance is insufficient to cover the requested withdrawal amount, the application should inform the user and terminate the transaction.



Questions??

- What is Activity Diagram? Explain all notations with examples.
- Draw activity diagram for login use case considering different scenarios.[D- 14]

Construct Activity diagrams for:

- Activities involved in ATM transaction
- Deposit Funds
- Withdraw Funds.